

Mapping the Intellectual Evolution of Degrowth Research

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Abstract

Degrowth has become one of the fastest-growing fields of sustainability research, yet its intellectual development has never been systematically reconstructed. Using a corpus of 2,927 peer-reviewed publications published between 1968 and 2025, we combine co-citation, co-word, and bibliographic coupling analyses to map the field's knowledge base, conceptual evolution, and emerging research front. Our analysis reveals that the intellectual centre of gravity of degrowth has shifted from articulating the theoretical foundations of degrowth toward empirically contesting green growth. The field's knowledge base is organised around three interconnected traditions—ecological economics, political ecology and post-development, and applied post-growth research paired with the critique of green growth—with the latter becoming the dominant intellectual hub in recent years. At the same time, the conceptual space of degrowth has expanded from a relatively concentrated critique of economic growth into a broad interdisciplinary research programme encompassing energy transitions, environmental justice, urban transformation, post-development, and alternative economic institutions. The emerging research front has likewise diversified toward post-growth macroeconomic modelling, decolonial perspectives, post-growth business, tourism, and sector-specific applications. By providing the first comprehensive bibliometric reconstruction of the field, this study offers a quantitative account of how degrowth has evolved and identifies promising directions for future research.

Keywords: Degrowth; ecological economics; sustainable development; political ecology; ecological macroeconomics; bibliometric analysis.

JEL Classification: B55, O44, P48, Q01, Q56, Q57

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1 Introduction

Over the past two decades, degrowth has evolved from a provocative slogan into a structured field of academic inquiry. Emerging from ecological economics and political ecology, and building on earlier critiques of indefinite economic expansion and the biophysical limits of growth (Georgescu-Roegen, 1971; Meadows et al., 1972; Daly, 1996), degrowth challenges the growth-centred economic model by advocating a democratically deliberated reduction in production and consumption, a more equitable distribution of resources, and a reorientation of social priorities toward well-being and ecological sustainability (Kallis, 2018; Hickel, 2020). The literature has since moved well beyond these disciplinary origins into environmental science, sociology, geography, business, planning, public policy, and the humanities, producing a large and increasingly diverse body of scholarship. This paper provides the first comprehensive bibliometric reconstruction of this evolving field.

The rapid expansion of this literature has transformed degrowth into one of the most dynamic areas of sustainability research. Hundreds of new publications now appear each year, generating a body of scholarship that is increasingly large, interdisciplinary, and conceptually diverse. This growth has enriched the field, but it has also made its overall structure more difficult to apprehend. No individual researcher can easily reconstruct, through close reading alone, the intellectual foundations of degrowth, the relationships among its different theoretical traditions, or the evolution of its research themes over time. Important questions therefore remain difficult to answer using conventional approaches: Which publications constitute the intellectual foundations of degrowth? How have different traditions interacted and evolved? Which concepts have become central, which have declined in prominence, and which new themes are emerging? More broadly, where is the research frontier of degrowth scholarship moving?

These questions have become increasingly relevant as the field has matured and as a growing body of scholarship has sought to synthesize its development. Narrative reviews have examined degrowth as an emerging academic paradigm (Weiss and Cattaneo, 2017), synthesized its theoretical foundations and major debates (Kallis et al., 2018), and analysed the policy proposals associated with degrowth transformations (Cosme et al., 2017). More recent systematic reviews have screened and thematically coded subsets of the literature to identify dominant themes, research streams, and conceptual developments (Engler et al., 2024; Hanaček et al., 2020), while a smaller-scale citation network analysis mapped relationships among 179 publications spanning 2008–2016 (Vandeventer et al., 2019). Together, these studies have substantially advanced our understanding of degrowth scholarship by clarifying its principal debates, concepts, and areas of application. Yet important questions remain about the field’s large-scale intellectual organisation and evolution. In particular, we still lack a comprehensive reconstruction of how the knowledge base has developed, how the conceptual vocabulary has evolved over time, and how new research fronts have emerged across the literature.

To address this gap, we examine four research questions. First, what publications and intellectual traditions constitute the knowledge base of degrowth research? Second, what themes define the conceptual space of the field, and how has this space evolved over time? Third, which emerging areas characterize the contemporary research frontier of degrowth scholarship? Fourth, what gaps and future research directions become visible from the structure of the field itself? By answering these questions, we move beyond identifying influential publications or recurring topics to reconstruct the evolving architecture of degrowth as an interdisciplinary research programme.

To answer these questions, we use bibliometric analysis, a family of quantitative methods designed

to identify structures and patterns within large bodies of scientific literature (Donthu et al., 2021). We construct a database of 2,927 peer-reviewed publications retrieved from OpenAlex, including complete metadata and reference lists. We combine three complementary science-mapping techniques: co-citation analysis to reconstruct the knowledge base and intellectual structure of degrowth research, co-word analysis to identify the conceptual space of the field and trace its evolution over time, and bibliographic coupling to detect emerging research fronts (Zupic and Čater, 2015). Together, these approaches provide a comprehensive reconstruction of both the historical foundations of degrowth scholarship and its recent transformation.

Our findings reveal that degrowth scholarship has undergone a profound intellectual reconfiguration rather than a simple expansion in publication volume. First, the field's knowledge base is organized around three enduring co-citation traditions: the foundational articulation of degrowth and its ecological-economic underpinnings, political ecology and post-development thought, and applied post-growth research paired with the critique of green growth. These traditions provide the intellectual foundations upon which the field has developed, but their relative prominence has changed substantially over time. The early literature was anchored in ecological economics, biophysical limits, and critiques of growth, whereas more recent scholarship has increasingly concentrated on empirically assessing the compatibility between economic growth, ecological constraints, and sustainability transitions, alongside a growing effort to operationalise post-growth alternatives in policy and practice. This applied strand, initially absent from the field's intellectual structure, has emerged as the dominant contemporary cluster,¹ signalling a shift in the centre of gravity of degrowth research from defining the limits of growth toward evaluating the feasibility of growth-compatible sustainability pathways.

Second, our analysis shows that this intellectual transformation has been accompanied by a broad expansion of the field's conceptual and empirical scope. The term degrowth, as captured by our keyword-defined corpus, evolved from a marginal concept associated primarily with biology before 2000 into a mature interdisciplinary discourse spanning climate and energy, justice, urban transformation, innovation, post-development, and alternative economic institutions. At the same time, the emerging research frontier has become increasingly differentiated. Recent scholarship no longer focuses exclusively on articulating critiques of growth but increasingly engages with their practical implications through post-growth macroeconomic modelling, provisioning systems, post-growth business, energy transitions, tourism, and decolonial perspectives.

Taken together, these findings show that degrowth has evolved along three interconnected dimensions: from conceptual consolidation toward thematic diversification, from foundational critique toward empirical and policy-oriented engagement, and from a predominantly European intellectual tradition toward a more globally engaged and decolonially attentive research programme.

Related Literature. This paper contributes to existing degrowth scholarship in three main ways. First, it provides the first comprehensive bibliometric reconstruction of the intellectual structure of the field. Existing scholarship has advanced understanding of degrowth through conceptual syntheses, systematic reviews, and more recently a small-scale citation network analysis. Narrative accounts documented the emergence of degrowth as an academic paradigm and its principal debates (Weiss and Cattaneo, 2017;

¹ Green growth refers to the approach that seeks to reconcile continued economic growth with environmental sustainability by improving resource efficiency, reducing environmental impacts, and promoting technological innovation. It differs from degrowth in that it does not question the desirability of aggregate economic growth itself.

Kallis et al., 2018; Cosme et al., 2017). Subsequent systematic reviews synthesized selected bodies of literature to identify dominant themes and research trajectories (Hanaček et al., 2020; Engler et al., 2024). Vandeventer et al. (2019) provided an initial quantitative overview of the field through a citation network analysis of 179 publications spanning 2008–2016. Our contribution builds on and substantially extends this literature by analysing 2,927 peer-reviewed publications spanning 1968–2025 through three complementary science-mapping techniques—co-citation, co-word analysis, and bibliographic coupling—to reconstruct the field’s intellectual structure, conceptual evolution, and emerging research front.

Second, we uncover how the intellectual centre of gravity of degrowth research has shifted over time. Previous scholarship has traced the emergence of degrowth from ecological economics, biophysical limits, and critiques of growth-oriented development (Georgescu-Roegen, 1971; Meadows et al., 1972; Daly, 1996; Latouche, 2009). Our analysis extends this literature by showing how these foundational concerns have evolved within the expanding research programme. While ecological-economic critiques provided the initial intellectual foundation of degrowth, contemporary scholarship is increasingly organized around empirical examinations of green growth, decoupling, sustainability transitions, and the institutional challenges of alternative pathways (Kallis, 2011; Hickel and Kallis, 2020). Rather than simply identifying different strands of degrowth thought, we document their changing prominence and the field’s movement from articulating critiques of growth toward investigating post-growth transformations.

Third, we identify the emerging research frontier and unresolved directions of degrowth scholarship. Existing reviews have highlighted the increasing diversity of the field and its expansion toward issues of justice, governance, policy, and socio-technical transitions (Kallis et al., 2018; Hanaček et al., 2020; Engler et al., 2024). Our bibliographic coupling analysis advances this discussion by systematically identifying the research niches gaining prominence, including post-growth macroeconomic modelling, decolonial perspectives, urban degrowth, energy transitions, tourism, and post-growth business. It also reveals areas that remain comparatively less developed, particularly questions related to governance, institutional change, political feasibility, and the distributive consequences of transition. By mapping both the expansion and remaining gaps of the field, our study provides a macro-level account of the past evolution, present structure, and future trajectory of degrowth research.

The remainder of the paper is organized as follows. Section 2 presents the data and bibliometric methodology, including the construction of the publication database and the science-mapping techniques employed. Section 3 reports the results of the co-citation, co-word, and bibliographic coupling analyses. Section 4 discusses the implications of these findings for the evolution of degrowth research and identifies directions for future inquiry. Section 5 concludes.

2 Data and Methodology

Systematic and narrative reviews provide an in-depth, qualitative reading of a research field, but they typically cover only a limited number of studies (often a few dozen to a few hundred) and concentrate on narrow, discipline-specific questions (Snyder, 2019). They are therefore ill-suited to charting a field as large and dispersed as degrowth in its entirety. Bibliometric analysis is the complementary instrument for this task: with its macro focus, broad timespan, and capacity to handle thousands of documents at once, it yields a holistic, relational map of a research domain (its knowledge base, conceptual structure,

and emerging frontiers), revealing patterns of which even domain experts may be unaware (Donthu et al., 2021), rather than a selective picture of a single topic or theme.

Our approach combines a carefully assembled corpus with a set of complementary mapping techniques. We first describe how the degrowth corpus was drawn from OpenAlex and profile its main features, and then introduce the three science-mapping techniques used to chart the field’s intellectual structure.

2.1 Data

We retrieved our bibliographic data from OpenAlex, an open and comprehensive scholarly database that indexes works across all disciplines and provides full reference lists for citation-based analyses. We use OpenAlex because, unlike the proprietary Web of Science and Scopus, it is freely and openly available, permitting fully reproducible bibliometric studies. Recent benchmarking further shows that, for publications indexed across all three databases, OpenAlex provides source-reference coverage comparable to Web of Science and Scopus (Culbert et al., 2025), making it a reliable platform for citation-based analyses. We queried all works whose title or abstract contained the term *degrowth*, retrieved through the OpenAlex API in 2026. The initial set was restricted to peer-reviewed scholarly document types (journal articles, reviews, books, and book chapters) published up to and including 2025. After excluding records hosted on preprint servers, institutional repositories, and thesis collections (which are not peer-reviewed), removing retracted and paratextual records, and deduplicating by identifier, DOI, and title,² this yielded a final corpus of 2,927 documents spanning 1968–2025. No language restriction was applied, so that the field’s substantial French, Spanish, and German literatures are represented. For each document we extracted the title, abstract, authorship and affiliation information, source, publication year, citation counts, and cited references.

Before turning to the bibliometric methods, we describe the resulting corpus along several standard dimensions: its growth over time, disciplinary composition, leading outlets, most productive authors, and geographic distribution.

Figure 1a depicts the evolution of publication activity over the period 1968–2025. Before 2010, the number of documents remained negligible, indicating limited scholarly engagement with the topic during this phase. Output then climbs steadily, from around forty documents in 2011 to nearly two hundred by 2019, as the topic attracts growing interdisciplinary interest. Growth accelerates further in the most recent years (apart from a brief dip in 2022), rising to a peak of 507 documents in 2025. Overall, the trajectory shows a clear shift, from marginal attention to sustained, intensive research, the same steep rise that recent reviews of the field have charted (Kallis et al., 2018; Engler et al., 2024).

The composition of this literature is as revealing as its volume. Figure 1b presents the disciplinary distribution of degrowth publications as a share of the corpus. The Social Sciences account for the largest share at 39.3%, followed by Environmental Science at 18.6%. Economics, Econometrics and

²Like all large bibliographic databases, including proprietary alternatives, OpenAlex may contain duplicate records, making record consolidation a standard preprocessing step in bibliometric research. We removed exact and near-duplicate documents by identifier, DOI, normalised title, publication year, and first author, together with the few remaining records sharing identical full titles, eliminating 118 duplicate documents. For the co-citation and bibliographic-coupling analyses, we likewise merged duplicate reference records into a single canonical identifier, chosen as the variant with the most complete bibliographic metadata (typically also the most frequently cited), with every merge verified manually. This prevents citations from being fragmented across duplicate records and understating a work’s influence.

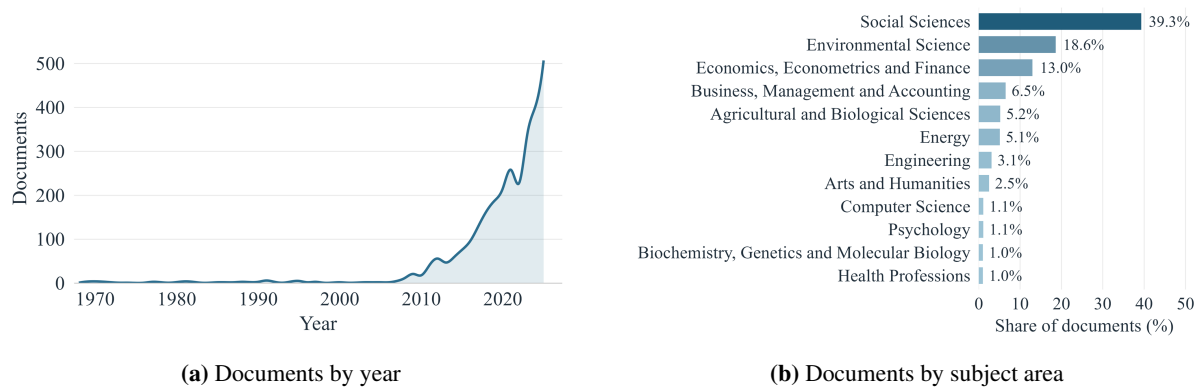


Figure 1: Growth and disciplinary composition of the corpus

Notes: Panel (a) plots the annual number of degrowth publications, 1968–2025 ($N = 2,927$). Panel (b) shows the disciplinary distribution as a share of all 2,927 documents, based on each work’s primary field in the OpenAlex classification.

Finance contribute 13.0% and Business, Management and Accounting 6.5%, while the Agricultural and Biological Sciences (5.2%) and Energy (5.1%) follow closely. Engineering (3.1%) and Arts and Humanities (2.5%) account for smaller shares, with the remainder distributed across Earth and Planetary Sciences, Computer Science, and other fields. This distribution indicates that degrowth research is predominantly anchored in the Social Sciences and Environmental Science, the interdisciplinary breadth long stressed in stock-takings of the field (Weiss and Cattaneo, 2017). The strong presence of Economics and Business signals a sustained focus on policy and managerial dimensions, while contributions from Energy, Engineering, and the Agricultural and Biological Sciences highlight the technological, resource-oriented, and biophysical perspectives that the field integrates.

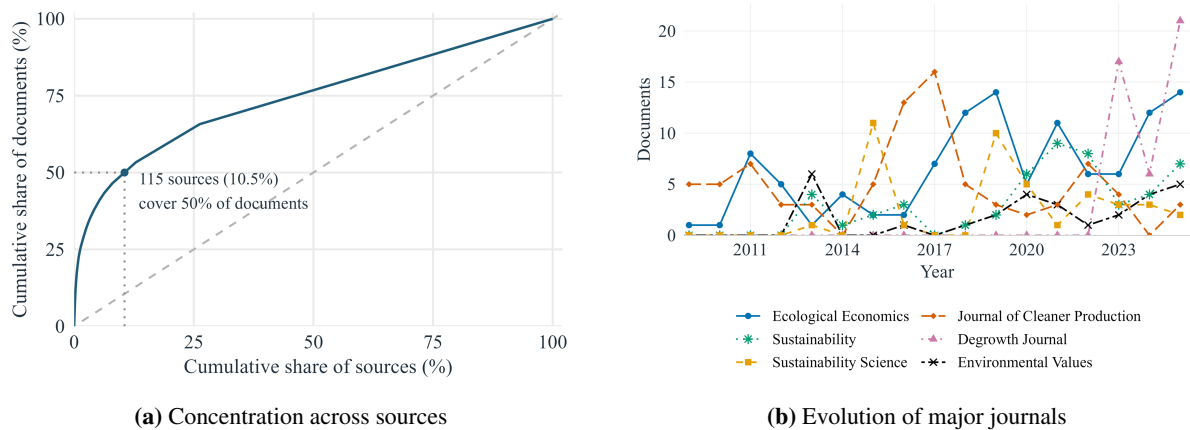


Figure 2: Where degrowth is published

Notes: Panel (a) plots the cumulative share of documents against the cumulative share of sources, ranked from most to least productive (a Lorenz curve); the dashed diagonal marks a perfectly even spread. Panel (b) plots the annual number of degrowth publications, 2008–2025, for the six most productive peer-reviewed journals (book series and preprint repositories excluded).

That interdisciplinary breadth is reflected in where degrowth research is published. Although *Ecological Economics* is the single largest outlet, accounting for 3.8% of the corpus and reflecting the field’s roots in ecological economics (Martínez-Alier et al., 2010), publication is remarkably dispersed. The 2,927 documents are distributed across roughly 1,100 distinct sources, nearly three-quarters of which publish only a single degrowth paper. Figure 2a makes this dispersion explicit: the most productive

10% of sources account for only half of all publications, meaning that more than one hundred different venues are needed to reach 50% of the corpus. Degrowth research is therefore not concentrated in a handful of flagship journals but diffuses across a wide range of disciplinary outlets, reflecting its strongly interdisciplinary character.

Even within this highly dispersed publication landscape, a small group of journals provides a recurring forum for the field. Figure 2b traces the annual output of the six most productive peer-reviewed journals—*Ecological Economics*, the *Journal of Cleaner Production*, *Sustainability*, the *Degrowth Journal*, *Sustainability Science*, and *Environmental Values*—which together account for about 12% of the corpus (roughly one-sixth of all journal publications). *Ecological Economics* maintains a steady presence throughout the period, consistent with its role in publishing foundational contributions such as [Martínez-Alier et al. \(2010\)](#) and [Kallis \(2011\)](#). The *Journal of Cleaner Production* expands rapidly before peaking around 2017 and subsequently declining, whereas the most striking development is the emergence of the dedicated *Degrowth Journal*. Since its launch in 2023, it has become the field’s most active publication venue by 2025, signalling the growing institutional consolidation of degrowth as a distinct area of scholarship. The remaining journals contribute more intermittently, reflecting the continued diffusion of degrowth research across the broader sustainability literature.

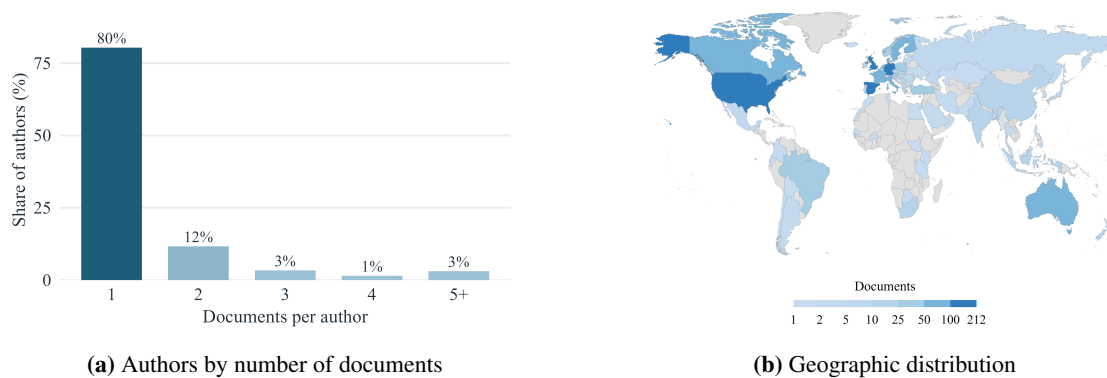


Figure 3: Authorship and geography of the corpus

Notes: Panel (a) shows the share of the 3,727 distinct authors by their number of degrowth papers (each author counted across all co-authored documents). Panel (b) maps the number of documents by corresponding-author country, shaded by graduated count classes; countries with zero degrowth documents in the corpus are shown in grey.

Authorship is similarly uneven. Figure 3a shows that the overwhelming majority of contributors engage with degrowth only once: four-fifths of the 3,727 distinct authors appear on a single paper, a further 12% on two, and only about 3% on five or more. The field thus rests on a small, recurring core of scholars surrounded by a broad, shifting periphery of occasional contributors, the mark of a young field, recently characterised by [Kallis et al. \(2018\)](#), that is still drawing in newcomers faster than it accumulates specialists.

The geographic distribution of the literature is similarly uneven. Figure 3b maps publications by corresponding-author country and reveals a pronounced concentration in Europe. The United Kingdom leads with 212 publications, followed by the United States (163), Spain (162), and Germany (150). A second tier of European countries, together with Australia and Canada, also contributes substantially, whereas output across much of the Global South remains comparatively limited.³ This pattern reflects the predominantly European origins of degrowth as an intellectual and political project ([Demaria et al.,](#)

³This proxy captures the institutional location of researchers rather than the populations or places their scholarship exam-

2013), while lending empirical support to recent calls for broader geographical representation within the field (Hanaček et al., 2020).

Taken together, these descriptive indicators portray a field that has expanded rapidly while undergoing institutional consolidation. Degrowth research is interdisciplinary in scope, widely dispersed across publication outlets despite the emergence of dedicated journals, and still geographically concentrated in Europe and North America. However, descriptive statistics alone cannot reveal how the field is intellectually organised. They do not identify the works that constitute its knowledge base, show how its conceptual vocabulary has evolved, or reveal the themes shaping its contemporary research frontier. Addressing these questions requires science-mapping techniques that reconstruct the relationships among publications, concepts, and citations. We now turn to those methods.

2.2 Methodology

Science mapping examines the relationships among research constituents—documents, references, and terms—to reveal the intellectual and conceptual organisation of a scientific field (Donthu et al., 2021; Zupic and Čater, 2015). We employ three complementary techniques, each addressing a distinct research question: co-citation analysis reconstructs the field’s knowledge base and intellectual structure; co-word analysis maps its conceptual space and evolution over time; and bibliographic coupling identifies its recent developments and emerging research fronts.

All three techniques represent the literature as a network: nodes joined by weighted links. What changes from one technique to the next is the unit of analysis and the meaning of a link. In every case, nodes that link strongly and often gather into densely connected groups, or clusters, which we read as the field’s main research streams. The three techniques look at degrowth through three lenses: its foundations, its vocabulary, and its frontier.

Co-citation analysis looks at the works that degrowth publications cite. Two earlier works are co-cited when a later document lists both in its reference list, and the more often two works appear together, the stronger we draw the link between them. Authors cite jointly the references they treat as part of the same intellectual tradition, so clusters of co-cited works expose the field’s knowledge base, the foundational texts it builds on. The 2,927 corpus documents together make some 87,000 citations to roughly 51,000 distinct works; mapping all of these would yield an unreadable network dominated by once-cited references, so we retain only the ninety most-cited, the field’s core intellectual foundations,⁴ and link each pair by the number of corpus documents that cite both. The network is thus built from the citation behaviour of the entire corpus, even though the map itself shows only its most influential references. Formally, let A be the document–reference incidence matrix, with $A_{di} = 1$ when document d cites reference i and $A_{di} = 0$ otherwise. The co-citation weight between references i and j is the number of documents that cite both,

$$C_{ij} = \sum_d A_{di} A_{dj}, \quad C_{ii} \equiv 0, \quad (2.1)$$

and these weights C_{ij} define the links of the co-citation network.

ines. This distinction is particularly important for the decolonial-degrowth research identified later in the coupling analysis (Section 3.3), where a European-based scholar studying degrowth in the Global South is classified here as European.

⁴We retain the ninety most-cited references because, among the cutoffs we tested, ninety gives the clearest and most interpretable cluster structure; the supporting sensitivity analysis is reported in Section 3.4.

Co-word analysis looks at the language of the publications themselves rather than at their references. Two terms co-occur when they appear in the same title or abstract, and we link frequently co-occurring terms more strongly; clusters of such terms map the field’s conceptual space, its leading themes. Letting A now denote the document–term incidence matrix, the weight between two terms is given by the same expression, $C_{ij} = \sum_d A_{di} A_{dj}$, the number of documents in which both occur. To follow how this space moved, we built a separate network for each of four periods (1968–2000, 2001–2010, 2011–2020, and 2021–2025).⁵ We extracted the terms from titles and abstracts: we lower-cased the text, removed stop-words and generic academic vocabulary (including the word *degrowth* itself, which would otherwise swamp every map), and dropped very short tokens. For the two sparse early periods we used single words and two-word phrases; for the dense recent periods we used two-word phrases only, which name more specific concepts such as climate change or ecological economics. Within each period we kept the terms that occur at least a minimum number of times (three, four, twelve, and eighteen, respectively, as the volume of text grows).⁶

Bibliographic coupling mirrors co-citation. Two documents are coupled when they share one or more references, and the more references they share, the stronger the link; with A the document–reference incidence matrix, the coupling weight between documents d and e is

$$B_{de} = \sum_i A_{di} A_{ei}, \quad B_{dd} \equiv 0. \quad (2.2)$$

Co-citation reflects the choices of citing authors and builds up slowly, since a work must be cited many times to enter the knowledge base; coupling is set the moment a paper is published, so it captures the field’s research front, groups of recent documents that draw on a common literature before they themselves are widely cited. We built two coupling networks, for 2011–2020 and 2021–2025. Older documents have had longer to gather citations, so we set a higher entry threshold for the first decade (at least thirty citations) than for the recent one (at least twelve), which leaves the two networks of comparable size (246 and 243 documents).

The path from a network to its clusters is the same for all three techniques. Write w_{ij} for a link weight (a C_{ij} or B_{de} above), $s_i = \sum_j w_{ij}$ for the strength of node i , and $m = \frac{1}{2} \sum_i s_i$ for the total link weight. Raw counts reward frequent nodes simply for being frequent, so we gauge the relative strength of each link by its association strength (van Eck and Waltman, 2009),

$$a_{ij} = \frac{2m w_{ij}}{s_i s_j}, \quad (2.3)$$

which compares the observed weight between two nodes with the value their strengths would predict; this normalised strength also drives the network layout. We then partition each network with the Leiden

⁵The periods follow decade boundaries from 2001 onward, both for comparability and because they track the field’s development: the emergence of degrowth as an academic paradigm during the 2000s, its consolidation through the 2010s, and its recent acceleration. The first period is left longer (1968–2000) because output was negligible before then (Figure 1a) and too sparse to subdivide; it captures the pre-degrowth era in which the term circulated mainly outside the social sciences.

⁶These thresholds are calibrated for legibility and rise with the volume of text in each period: a low cutoff (three or four occurrences) is needed in the sparse early periods to retain enough recurrent terms to form a network at all, whereas the dense recent periods require a higher cutoff (twelve, then eighteen) to filter out idiosyncratic vocabulary and keep the map interpretable.

algorithm (Traag et al., 2019), which assigns each node to a cluster g_i so as to maximise the modularity

$$Q = \frac{1}{2m} \sum_{i,j} \left(w_{ij} - \gamma \frac{s_i s_j}{2m} \right) \delta(g_i, g_j), \quad (2.4)$$

where $\delta(g_i, g_j) = 1$ when i and j lie in the same cluster and 0 otherwise. The resolution γ sets how finely the network is divided; we tuned it to return a small, interpretable set of clusters, and we fixed the random seed so the partition reproduces exactly. Some nodes end up with no links to the rest of the network, or form small groups linked only to one another; we discarded these and kept only each network’s main connected component, the largest group of nodes that are all reachable through links.

Finally, we laid out each network with a force-directed algorithm that pulls strongly linked nodes together and pushes weakly linked ones apart, while giving within-cluster links extra weight so that clusters occupy distinct regions of the layout. Colour indicates cluster membership, and node size represents prominence: the number of co-citations for references, the frequency of terms in the corpus for co-word networks, and total coupling strength for bibliographic-coupling networks. We label only the most prominent nodes in each cluster.⁷

3 Results

We report the results of the three science-mapping analyses, which take up our first three research questions in turn: co-citation analysis recovers the field’s knowledge base, co-word analysis maps its conceptual space and how it has evolved, and bibliographic coupling traces its emerging research front.

3.1 Intellectual Structure: Co-Citation Analysis

To investigate the intellectual structure of the degrowth literature, we performed a co-citation analysis of the references cited by the corpus. The underlying premise of co-citation analysis is that documents frequently cited in tandem are likely to share conceptual or methodological affinities and thus belong to the same research tradition or thematic domain. From the references cited across the corpus, the analysis retained the ninety most-cited as the nodes of the network, capturing the field’s most influential contributions.⁸ The co-citation network was normalised using association strength and partitioned with the Leiden community-detection algorithm, which groups references by maximising internal co-citation strength.

Each resulting cluster represents a coherent body of literature, delineated by shared intellectual concerns and citation patterns. The network (Figure 4) resolves into three clusters, each corresponding to a major thematic orientation within the degrowth research landscape: the foundational articulation of degrowth and its ecological-economic underpinnings; political ecology and post-development thought;

⁷For visualisation, node radii are scaled so that node area is proportional to prominence. Specifically, if p_i denotes node prominence, the displayed radius is $r_i = r_{\min} + (r_{\max} - r_{\min}) \sqrt{(p_i - p_{\min}) / (p_{\max} - p_{\min})}$, where $r_{\min}$ and $r_{\max}$ are the smallest and largest displayed node radii and $p_{\min}$ and $p_{\max}$ are the minimum and maximum prominence values in the network.

⁸Node labels in Figure 4 and the years in Table 1 report the publication year that OpenAlex records for each reference. Because variant records of the same work were merged onto a single canonical record (Section 2), and because OpenAlex sometimes dates a work by its online rather than its print appearance, a label may differ by a year from the work’s commonly cited date; for instance, Hickel and Kallis’s “Is Green Growth Possible?” is labelled 2019 in the figure but cited as 2020 in the text and reference list.

and applied post-growth research and the critique of green growth (Table 1 lists the most co-cited works in each cluster). Read together, these three clusters give empirical shape to the intellectual currents that earlier, narrative accounts of the field had distinguished only qualitatively (Demaria et al., 2013).

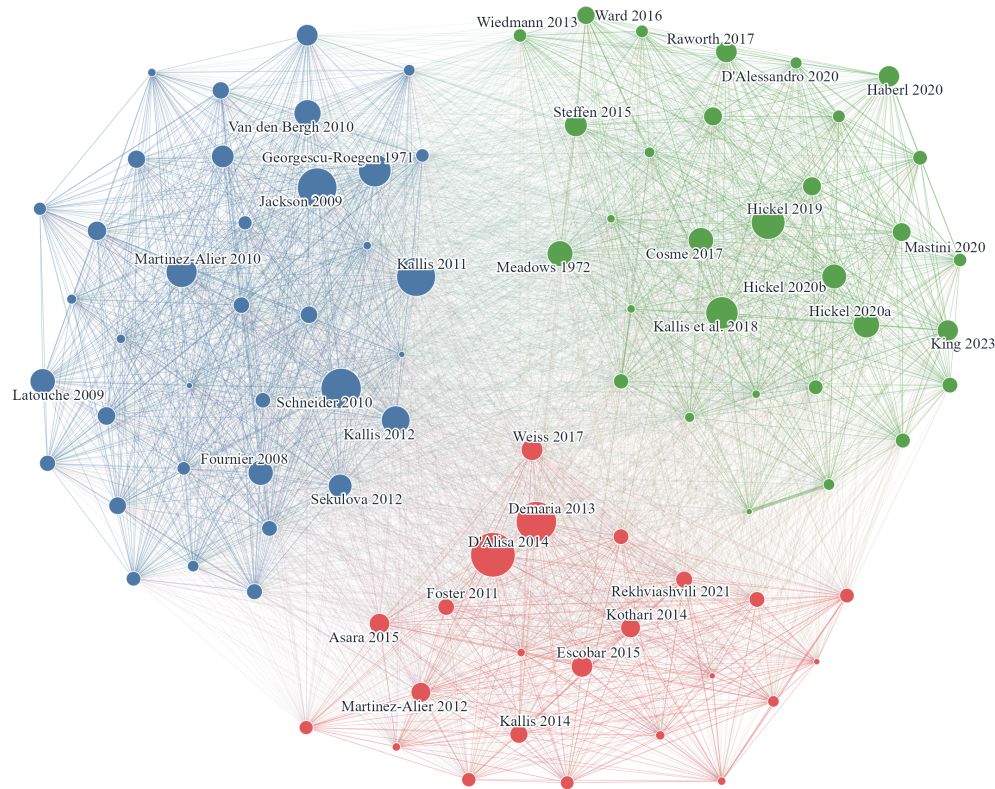


Figure 4: Co-citation network (1968–2025)

Notes: This figure displays the co-citation network of the ninety most co-cited references in the degrowth literature. Clusters are identified with the Leiden algorithm; node size is proportional to the number of co-citations.

Cluster 1: Foundational Degrowth Theory and Ecological Economics (Blue). This is the field’s foundational core, a broad and heterogeneous cluster merging the seminal articles that defined degrowth with the ecological-economic, biophysical, and steady-state classics on which they build and the early empirical work that put them into practice. Its most co-cited works are [Schneider et al. \(2010\)](#)’s framing of degrowth for ecological sustainability and social equity, [Jackson \(2009\)](#)’s *Prosperity Without Growth*, and [Kallis \(2011\)](#)’s “In Defence of Degrowth,” together with [Kallis et al. \(2012\)](#)’s statement of the economics of degrowth, [Georgescu-Roegen \(1971\)](#)’s entropic critique of the economic process, [Martínez-Alier et al. \(2010\)](#)’s mapping of degrowth within ecological economics, [van den Bergh \(2011\)](#)’s critical rejoinder proposing “a-growth” as an alternative to degrowth, and [Victor \(2008\)](#)’s *Managing Without Growth*. It also gathers the *décroissance* roots of the movement ([Fournier, 2008](#)), its early empirical and grassroots strand ([Sekulova et al., 2012](#); [Johanisova et al., 2012](#)), and the link to the political economy of inequality through [Piketty \(2014\)](#)’s *Capital in the Twenty-First Century*. The cluster articulates degrowth as a multidimensional critique of growth-centred economics anchored in ecological-economic

Table 1: The knowledge base of degrowth: most co-cited references by cluster

Reference	Focus	Co-citations
<i>Cluster 1: Foundational degrowth theory and ecological economics (Blue)</i>		
Schneider et al. (2010)	Degrowth for sustainability and equity	287
Jackson (2009)	Post-GDP macroeconomics	276
Kallis (2011)	Case for degrowth as a political project	267
Georgescu-Roegen (1971)	Entropic critique of the economic process	187
Martínez-Alier et al. (2010)	Degrowth within ecological economics	170
Kallis et al. (2012)	Macroeconomics of non-growing economy	146
van den Bergh (2011)	Case for growth-agnostic “a-growth”	131
<i>Cluster 2: Political ecology, post-development, and the degrowth vocabulary (Red)</i>		
D’Alisa et al. (2014)	Glossary of core degrowth concepts	355
Demaria et al. (2013)	Origins and meanings of degrowth	289
Weiss and Cattaneo (2017)	Degrowth as an emerging paradigm	89
Escobar (2015)	Post-development and transitions	88
Asara et al. (2015)	Social-ecological transformation	81
Kothari et al. (2014)	Alternatives to sustainable development	77
Kallis and March (2014)	Degrowth as utopian imaginary	67
<i>Cluster 3: Applied post-growth research and the critique of green growth (Green)</i>		
Hickel and Kallis (2020)	Decoupling critique	200
Hickel (2020)	Post-growth programme	125
Cosme et al. (2017)	Degrowth policy proposals	117
Haberl et al. (2020)	Systematic review of decoupling evidence	86
Fitzpatrick et al. (2022)	Systematic mapping of degrowth policy	71
Ward et al. (2016)	Empirical test of decoupling feasibility	46
D’Alessandro et al. (2020)	Feasible policy alternatives to green growth	44

Notes: For Clusters 1–2 we list the seven most-cited works; ranking strictly by frequency would otherwise let the largest clusters crowd out the smallest. For Cluster 3 we list the most-cited work representing each of its principal sub-themes (empirical critique of green growth; applied/policy operationalisation of post-growth alternatives), since its most frequently cited works alone would not represent the cluster’s thematic range. “Co-citations” is the number of corpus documents citing the reference together with another retained reference. Cluster colours match the network map.

and steady-state thought.

Cluster 2: Political Ecology, Post-Development, and the Degrowth Vocabulary (Red). This cluster gathers texts rooted in political ecology, cultural critique, and post-development thought. Its most co-cited works are the vocabulary volume ([D’Alisa et al., 2014](#)) and [Demaria et al. \(2013\)](#)’s account of the origins and meanings of degrowth, followed by [Weiss and Cattaneo \(2017\)](#)’s stock-taking of the field and [Escobar \(2015\)](#)’s post-development critique. It also includes [Asara et al. \(2015\)](#)’s account of degrowth as a social–ecological transformation, [Kothari et al. \(2014\)](#)’s pairing of degrowth with *buen vivir* and ecological swaraj, and [Kallis and March \(2014\)](#)’s reading of degrowth as a utopian imaginary.

Cluster 3: Post-Growth and the Critique of Green Growth (Green). The most contemporary and fastest-growing cluster spans two related strands: an empirical critique of green growth and the applied operationalisation of post-growth alternatives. The critique strand is anchored by [Hickel and Kallis \(2020\)](#)’s “Is Green Growth Possible?”, alongside [Haberl et al. \(2020\)](#)’s systematic review of the evidence on decoupling and [Ward et al. \(2016\)](#)’s empirical test of whether GDP growth can be decoupled from environmental impact. The applied strand includes [Hickel \(2020\)](#)’s *Less Is More* and the systematic mappings of degrowth policy proposals by [Cosme et al. \(2017\)](#) and [Fitzpatrick et al. \(2022\)](#),

while [D’Alessandro et al. \(2020\)](#)’s modelling of feasible policy alternatives to green growth bridges both strands directly. This cluster frames degrowth’s contemporary research front as an empirically grounded challenge to green-growth narratives, paired with a growing effort to translate post-growth ideas into concrete policy proposals.

Although the map aggregates the full period, the knowledge base it depicts is not static. Recomputing co-citation period by period, holding the three clusters fixed and measuring each cluster’s share of co-citation among the documents of each period, shows the green-growth cluster rising from no presence before 2011 to 13% of co-citation mass in 2011–2020 and 49% in 2021–2025, overtaking the foundational core, whose share falls from 99% in 2001–2010 to 31%, to become the field’s centre of gravity (Appendix [A.2](#)). We return to this shift in Section [4](#).

3.2 Conceptual Structure: Co-Word Analysis

This section presents a co-word analysis of the degrowth literature, identifying and interpreting the semantic structures of research themes over four distinct periods: 1968–2000, 2001–2010, 2011–2020, and 2021–2025. Using term co-occurrence networks built from titles and abstracts, we examine how conceptual clusters emerged, evolved, and diversified over time, providing a detailed account of the shifting intellectual landscape within the field of degrowth. We read these maps at the level of broad thematic formations: the number and identity of the major clusters are stable across specifications, whereas the exact placement of an individual term is more sensitive to the frequency threshold and is therefore treated as indicative rather than definitive (Appendix [A.3](#)). Across the four periods analysed, the indexed degrowth literature has evolved from a marginal and biologically coloured discourse into a richly structured, interdisciplinary, and politically engaged field. From the foundational critiques of growth and sustainability, it has expanded into questions of governance, justice, post-capitalist systems, and ecological ethics. The increasing complexity and thematic depth of co-word clusters over time confirm a clear trajectory: from a scattered start to a mature, well-structured field.

1968–2000 in the English-Keyword Corpus: A Biological Register. In the earliest period (1968–2000), the co-word network (Figure [5](#)) reveals a sparse and fragmented structure, with three distinct clusters emerging under a minimal occurrence threshold of 3. This pattern reflects how little of degrowth’s socio-economic vocabulary had yet entered the indexed literature: loosely connected, and dominated by biological terms.

The network is dominated almost entirely by biological vocabulary, confirming that before the 2000s the term “degrowth” circulated mainly in a physiological rather than a social-scientific register. Three clusters emerge. The *red* cluster gathers terms such as freshwater, tissue, starvation, snails, and population, reflecting studies of regeneration and metabolism in freshwater invertebrates. The *blue* cluster (food, energy, body, oxygen, Gastropoda, water, and rates) corresponds to metabolic-rate and energy-budget research, while the *green* cluster (size, cell, planarian, Dugesia, regeneration) captures cellular-regeneration studies in flatworms. The normative, economic meaning of degrowth is essentially absent.

Taken together, these clusters show that before 2000 the term “degrowth” belonged primarily to a biological rather than a socio-economic vocabulary. Only a faint trace of its later normative meaning appears in the corpus. This should not be interpreted as evidence that the intellectual foundations of degrowth were absent. Seminal works such as [Georgescu-Roegen \(1971\)](#)’s entropic critique of economic

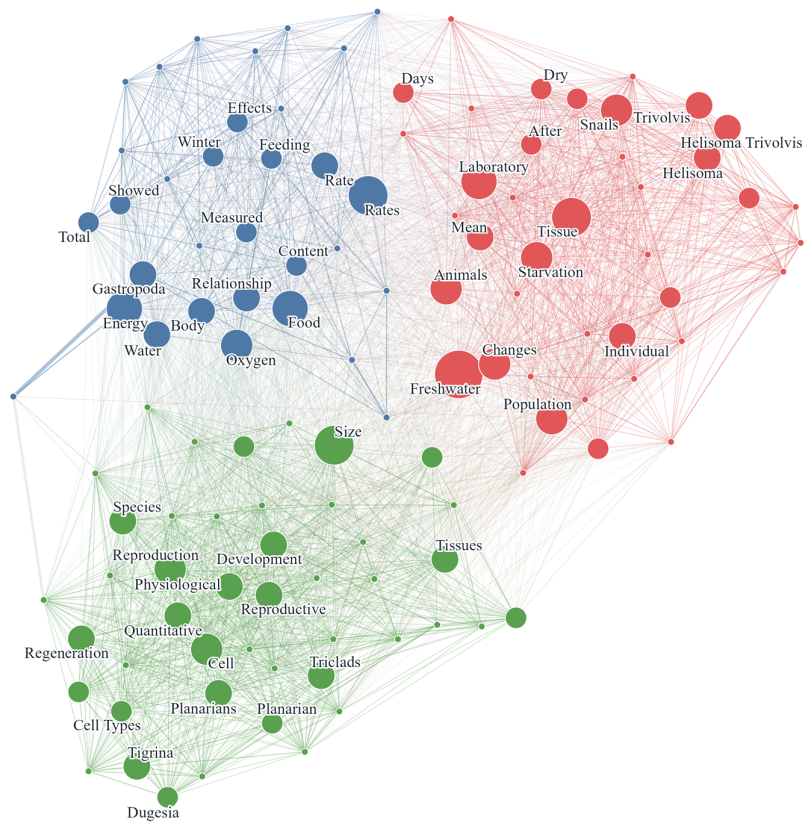


Figure 5: Co-word network (1968–2000)

Notes: This figure displays the co-word network, built from terms extracted from titles and abstracts; clusters are identified with the Leiden algorithm.

growth and [Daly \(1991\)](#)’s steady-state economics predate the emergence of degrowth as a research field and later become central to its co-citation knowledge base. They are absent from our corpus because neither uses the term *degrowth*, on which our keyword-based search is defined. The same limitation applies to the concept’s own socio-economic origins. Although *d’ecroissance* was introduced by [Grinevald and Rens \(1979\)](#), nearly a decade before the end of this period, that foundational French-language literature falls outside a corpus constructed from the English keyword *degrowth* (the resulting coverage gap is quantified in Appendix A.6). This period should therefore be interpreted not as the beginning of degrowth research, but as the final stage in which degrowth was predominantly a biological term. The socio-economic concept already existed in French, while its English-language intellectual antecedents were articulated under different terminology. Only later did the English term *degrowth* become associated with that socio-economic tradition.

2001–2010: Emergence of Degrowth as an Economic Paradigm. In the subsequent decade (2001–2010), the co-word structure captures the emergence of degrowth as a distinct research programme within ecological economics and the broader sustainability literature (Figure 6).

During this transitional decade, degrowth emerged as a coherent socio-economic paradigm while

matic formations that reflect a broadening interdisciplinary engagement and a widening of normative and analytical concerns.



Figure 7: Co-word network (2011–2020)

Notes: This figure displays the co-word network, built from multi-word terms extracted from titles and abstracts; clusters are identified with the Leiden algorithm.

Seven thematic formations structure this decade. The *red* cluster forms the climate-and-energy axis (climate change, renewable energy, green deal, planetary boundaries, greenhouse-gas emissions), drawing in steady-state and state-economy vocabulary. The *blue* cluster (sustainable development, natural resources, development goals, circular economy, tourism development) engages the sustainability and resource-governance agenda. The *green* cluster joins production and consumption, ecological sustainability, social justice, and anti-capitalist critique around an emerging economic paradigm. The *orange* cluster names the field's transformational core (social-ecological and socio-ecological transformation, socially sustainable degrowth). The *purple* cluster assembles the theoretical anchors of ecological economics, political economy, political ecology, and environmental justice. The *teal* cluster collects political-economic and crisis vocabulary (socio-economic, ecological crisis, civil society, basic income), while the *yellow* cluster foregrounds post-development and decolonial currents (global South and global North, *buen vivir*, social movements, green economy, and *del decrecimiento*).

In sum, the period 2011–2020 is marked by thematic diversification and intellectual consolidation. Degrowth is no longer merely a critique of economic growth but has matured into a multi-dimensional research programme that encompasses political economy, macroeconomic theory, policy design, ethics,

and post-developmental imaginaries. Biological associations have receded to the margins, while governance, justice, and systemic transformation have moved to the centre of scholarly inquiry. The maturation visible across this decade is just what Weiss and Cattaneo (2017) registered in taking stock of degrowth as an emerging academic paradigm.



Figure 8: Co-word network (2021–2025)

Notes: This figure displays the co-word network, built from multi-word terms extracted from titles and abstracts; clusters are identified with the Leiden algorithm.

2021–2025: Multipolar Convergence and Normative Synthesis. The most recent period is characterised by an unprecedented proliferation and diversification of research themes. The co-word network (Figure 8) resolves into ten thematic formations spanning energy, sustainability, justice, and post-development concerns. The *red* cluster is the field’s energy-and-climate axis (climate change, climate crisis, green deal, renewable energy, carbon emissions, energy transition, and fossil fuels), articulating the critique of market-based decarbonisation. The *blue* cluster gathers the sustainability and resource agenda (sustainable development, circular economy, natural resources, biodiversity loss, steady state, development goals), while the *green* cluster foregrounds political ecology and global justice (global South and global North, political ecology, environmental and climate justice, social movements).

The remaining clusters mark the field’s growing thematic breadth. The *purple* cluster, the largest, names the socio-ecological transformation and its recent eco-Marxist turn (political economy, socio-ecological and ecological transformation, ecological economics, with prominent terms on Marx, the

Anthropocene, and the work of Kohei Saito); the *orange* cluster links production and consumption to planetary boundaries and environmental impacts; and the *teal* cluster engages economic systems and models. The *yellow* cluster joins social justice, economic development, and post-development (*buen vivir*), while the *brown* cluster captures the pandemic-and-tourism turn (COVID-19 pandemic, global economy, sustainable tourism, tourism development). Finally, the *pink* cluster gathers alternative economic systems and the critique of capitalism (alternative economic models, capitalist system, environmental degradation), and the *grey* cluster points to an emerging urban and eco-social strand (urban planning, eco-social, climate emergency).

In sum, the 2021–2025 period marks the consolidation of degrowth as a heterogeneous, deeply interdisciplinary, and ethically expansive research programme. The literature has moved beyond critique to engage with implementation, policy, pedagogy, and post-capitalist visions, while remaining grounded in material and ecological realities, the breadth of a research programme that earlier syntheses had already begun to chart (Kallis, 2018).

3.3 Emerging Themes: Bibliographic Coupling

Bibliographic coupling provides a window onto the field’s emerging research frontier. Applied to the degrowth literature, it distinguishes two broad phases of development: a period of conceptual consolidation (2011–2020), followed by a phase of rapid thematic diversification and empirical expansion (2021–2025).

For each interval, the coupling network retains the most-cited documents of the period, using a citation threshold calibrated so that the two maps are of comparable size: documents with at least 30 citations for 2011–2020 and at least 12 citations for 2021–2025, yielding networks of 241 and 243 documents, respectively. Because earlier publications have had more time to accumulate citations, the higher threshold for the first decade prevents the network from becoming unwieldy. This calibration minimises noise and highlights the influential contributions shaping the trajectory of the field. As with the co-word maps, we interpret the coupling networks at the level of broad clusters rather than the exact assignment of individual documents, which is more sensitive to the citation threshold and the random seed and is read accordingly (Appendix A.4).

Our analysis reveals an increasingly differentiated research landscape. The early phase is characterised by the consolidation of core post-growth arguments and socio-ecological critiques, whereas the recent period exhibits a proliferation of thematic niches encompassing post-growth modelling, urban transformation, post-growth business and provisioning, decoloniality, and tourism. This evolution signals a shift from foundational debates toward applied, interdisciplinary engagements with systemic transitions.

Period 2011–2020: Foundation and Conceptual Consolidation. The 2011–2020 window represents a phase of intellectual consolidation. Eight clusters capture the core debates and conceptual architectures of this period.

Eight coupling clusters structure the 2011–2020 window (Figure 9). The *red* cluster, the largest, forms the field’s broad and undifferentiated early literature, spanning defining conceptual statements, applied case studies from Detroit to Cuba to Catalonia, and debates on democracy and consumption (Weiss and Cattaneo, 2017; Cosme et al., 2017; Kallis et al., 2012; Haapanen and Tapio, 2015). The

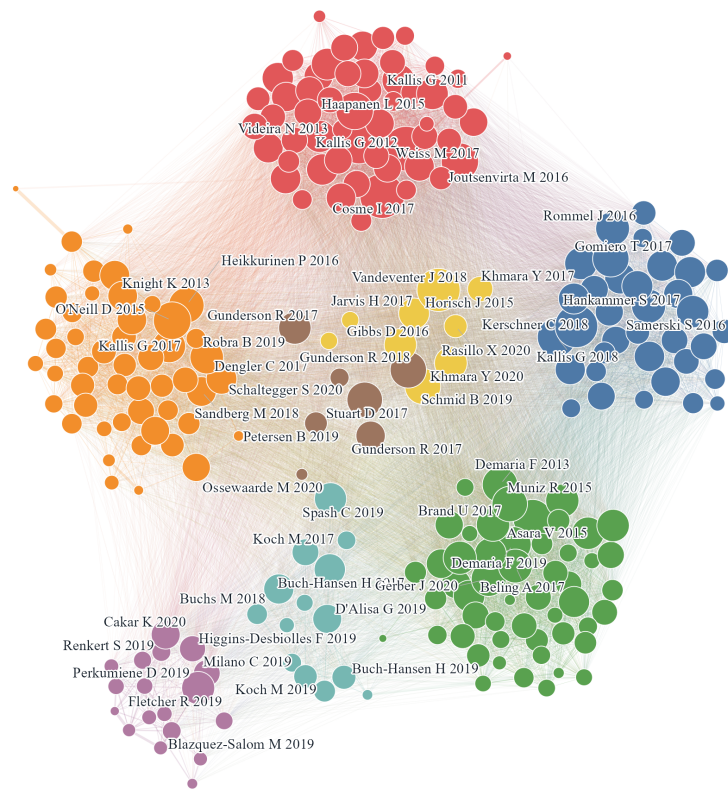


Figure 9: Overview of the emergent clusters (2011–2020)

Notes: This figure displays the bibliographic-coupling network of degrowth documents from 2011–2020 with at least 30 citations (241 documents). Clusters are identified with the Leiden algorithm; node size is proportional to coupling strength.

green cluster assembles political ecology, post-development, and decolonial scholarship (Asara et al., 2015; Demaria et al., 2019; Beling et al., 2017). The *orange* cluster gathers the empirical and theoretical debate over growth itself, spanning ecological macroeconomics and biophysical modelling, working-time research, and normative assessments of green growth versus degrowth (O’Neill, 2015; Althouse et al., 2020; Nieto et al., 2019), while the *blue* cluster collects work on innovation, provisioning, and business beyond growth (Kerschner et al., 2018; Khmara and Kronenberg, 2017; Kallis et al., 2018).

The remaining clusters mark early sectoral and institutional differentiation. The *teal* cluster consolidates the political economy, governance, and welfare of degrowth (Spash, 2019; Buch-Hansen, 2018; Buchs and Koch, 2019; D’Alisa and Kallis, 2019; Koch, 2019); the *purple* cluster develops a dedicated critique of tourism and urban growth (Fletcher et al., 2019; Milano et al., 2019); the *yellow* cluster addresses degrowth transitions and contrasts them with green-growth pathways for regional development (Vandeventer et al., 2019; Gibbs and O’Neill, 2016); and the *brown* cluster gathers a critical-theory reading of consumption and capitalism (Gunderson et al., 2018; Stuart et al., 2017). Together they reveal a centralised intellectual structure anchored in ecological economics and political ecology.

Period 2021–2025: Acceleration and Diversification. The 2021–2025 window reflects rapid thematic expansion and empirical deepening. Eleven clusters, ranging from 43 to 9 documents each, capture the

intellectual contours of this diversification.

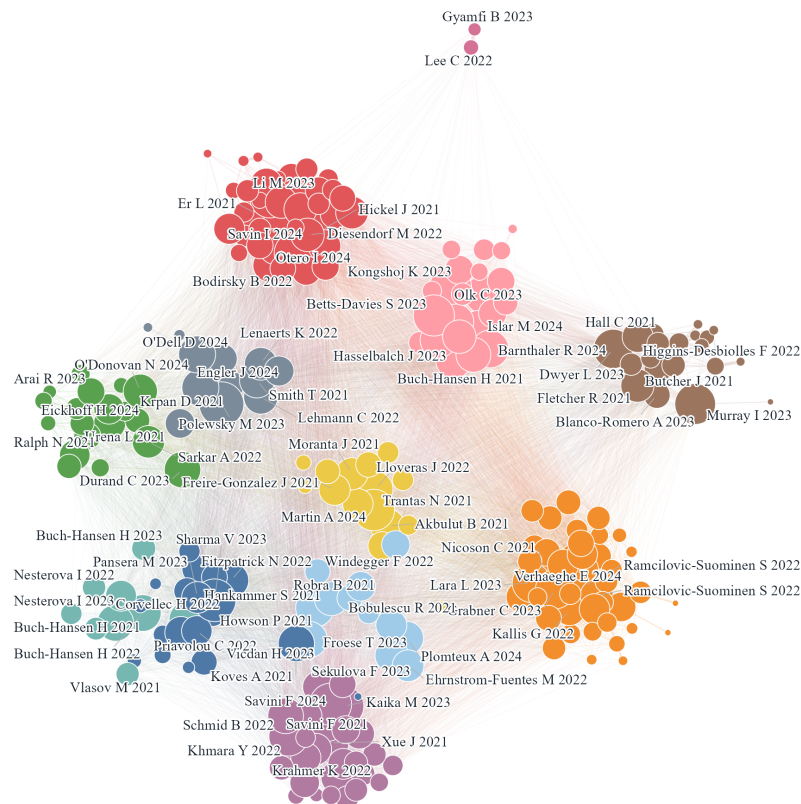


Figure 10: Overview of the emergent clusters (2021–2025)

Notes: This figure displays the bibliographic-coupling network of degrowth documents from 2021–2025 with at least 12 citations (243 documents). Clusters are identified with the Leiden algorithm; node size is proportional to coupling strength.

Eleven coupling clusters capture the diversification of the 2021–2025 window (Figure 10). The *orange* cluster, the largest, advances decolonial degrowth, political ecology, and global justice (Gräbner and Strunk, 2023; Guerrero Lara et al., 2023; Ramcilovic-Suominen et al., 2022). The *red* cluster models the macroeconomic and climate feasibility of post-growth trajectories (Otero et al., 2024; Li et al., 2023), and engages the debate over whether decarbonisation is achievable without such a transition (Diesendorf, 2022). The *purple* cluster concentrates urban degrowth and planning (Savini, 2024; Kaika et al., 2023; Khmara and Kronenberg, 2022; Xue, 2021). The *pink* cluster develops eco-social policy and the welfare state (Kongshøj, 2023; Schulze Waltrup, 2023; Buch-Hansen and Carstensen, 2021).

The remaining clusters mark the field's growing sectoral and thematic breadth. The *brown* cluster forms a dedicated tourism-degrowth subfield (Murray et al., 2023; Dwyer, 2023; Butcher, 2021); the *blue* cluster examines digital technology, infrastructure, and the commons (Sharma et al., 2023; Pansera et al., 2023); and the *green* cluster juxtaposes circular-economy and green-growth research against a critical, degrowth-informed alternative (Durand et al., 2023; Belmonte Ureña et al., 2021). The *yellow* cluster explores the discursive risks facing the term *degrowth* itself, alongside emerging work on needs, well-being, and sufficiency (Trantas, 2021; Helne, 2021); the *grey* cluster surveys public attitudes toward degrowth and green growth, alongside the field's emerging systematic reviews (Polewsky et al.,

2023; Engler et al., 2024); the *light-blue* cluster develops post-growth innovation and business models (Hankammer et al., 2021); and the *teal* cluster examines post-growth business, provisioning, and organisational change (Corvellec and Paulsson, 2022).

3.4 Robustness

We assess the robustness of our findings through a broad set of sensitivity analyses and validation exercises. Some tests examine whether the results depend on methodological choices, by varying one modelling decision at a time—including the reference cutoff, clustering algorithm and resolution, random seed, per-period thresholds, counting convention, and corpus boundary. Others address substantive concerns specific to our analyses, such as the temporal evolution of the co-citation structure, language coverage, and duplicate consolidation. The complete set of checks is reported in Appendix A; each paragraph below states what is varied and what the corresponding exercise demonstrates.

The three-cluster knowledge base recovered by the full-sample co-citation network is robust to every perturbation of the network construction (Appendix A.1). The three-cluster structure is preserved across reference cutoffs from the seventy to the one hundred and twenty most-cited nodes (adjusted Rand index 0.75 to 1.00 relative to the reported ninety-node network), coincides with the modularity-maximising resolution, is reproduced identically by the Louvain algorithm (Blondel et al., 2008), survives fifty random seeds (modal three clusters in forty-two of fifty; mean adjusted Rand index 0.91), and does not depend on our manual consolidation of duplicate references (adjusted Rand index 0.85). The recovered knowledge base is therefore a property of the data rather than of any particular modelling choice.

Table 2: Time-sliced co-citation: share of co-citation mass by knowledge-base cluster

Period	Political ecology	Foundational / eco-economics	Green-growth critique	Documents
1968–2000	–	–	–	0
2001–2010	1.8%	98.2%	0.0%	15
2011–2020	24.9%	63.1%	12.0%	84
2021–2025	25.7%	31.2%	43.1%	90

Notes: Each cell is cluster *c*’s share of total co-citation node strength among the ninety fixed reference nodes, computed from documents published in the period only; rows sum to 100% up to rounding. Clusters and colours are fixed from the full-corpus map (Figure 4). “Documents” counts the period’s corpus documents contributing at least one co-citation among the retained nodes. Internal density corroborates the same ranking: the green-growth cluster has the highest mean internal association strength (1.62, against 1.56 for political ecology and 1.33 for the foundational cluster).

Because the reported co-citation network pools the entire 1968–2025 period, it cannot reveal when each intellectual tradition became more or less prominent. To address this limitation, we recompute co-citation separately within each period while holding the three clusters fixed; Table 2 reports the results, and Appendix A.2 details the underlying method.¹⁰ The temporal pattern from 2001 onward is unambiguous: the foundational cluster accounts for almost the entire knowledge base in 2001–2010

¹⁰The 1968–2000 period records no co-citations among the tracked references because virtually no documents in our keyword-defined corpus were published before 2000 (Figure 1a). This does not imply that the intellectual foundations of degrowth were absent from the period: seminal works such as Georgescu-Roegen (1971) and Daly (1991) were already published and appear as reference nodes in the full-sample co-citation network. They acquire co-citations only once the later literature using the keyword *degrowth* begins citing them jointly.

(98.2% of co-citation mass) before declining to 31.2% by 2021–2025, while the applied post-growth and green-growth-critique cluster rises from no presence to 43.1%, becoming the field’s largest intellectual centre. This temporal decomposition confirms the centre-of-gravity shift that underpins the paper’s central finding.

The co-word networks are inherently more granular than the co-citation network, so we interpret them at the level of broad thematic formations rather than individual terms and assess their robustness to the term-frequency threshold and the random seed (Appendix A.3). In every period, the number and broad composition of clusters are preserved as the threshold varies around its reported value. The partitions are identical across random seeds in the two sparse early periods and remain modally stable in the denser recent periods. The exact position of individual terms is naturally more sensitive to these choices, but the thematic organisation of the maps remains unchanged.

The bibliographic-coupling networks, which map the field’s research frontier, are likewise tested for robustness to the citation-entry threshold and the random seed (Appendix A.4). Across these perturbations, the number and broad composition of clusters remain highly stable, with at most one additional or one fewer cluster under alternative specifications. The modal partition is also stable across random seeds, supporting interpretation at the level of broad thematic formations rather than the exact assignment of individual documents.

Also, the descriptive profile of the corpus could depend on how authors and countries are counted, so we re-estimate the two rankings most sensitive to these choices under alternative counting conventions (Appendix A.5). The geographic distribution is virtually unchanged whether publications are assigned by corresponding-author country or by whole counting across all co-author affiliations (Spearman rank correlation 0.95, with an identical top twelve), while the ranking of the most productive authors retains the same core under both full and fractional counting.

Because the corpus is defined by the presence of the English keyword *degrowth* rather than by the language of publication, it already captures non-English-language works that use this term; accordingly, French, Spanish, and other non-English expressions appear directly in the co-word maps. The remaining concern is different: scholarship published under cognate terms, such as *décroissance* or *decrecimiento*, but never using the English term may be missed. We assess the magnitude of this coverage gap using OpenAlex’s language-independent Degrowth concept (Appendix A.6). Among works identified by this concept but absent from our keyword-defined corpus because they rely exclusively on non-English terminology, we find only about sixty additional documents, representing a few percent of the corpus. This limited number suggests that the omission of cognate-language terminology is unlikely to alter the aggregate structure of the field.

The same keyword definition may also miss adjacent post-growth scholarship expressed through alternative terminology, such as post-growth or steady-state economics. We assess this boundary using OpenAlex’s Degrowth concept classification (Appendix A.7). Among the works assigned to this concept, 91% already contain the term *degrowth*, implying that a broader search based on related concepts would recover only about two hundred additional documents (nine percent of the corpus). Moreover, because co-citation analysis is based on cited references rather than document titles or abstracts alone, the intellectual foundations shared with adjacent post-growth traditions are already reflected in the knowledge-base analysis. These results suggest that the keyword definition captures the central structure of the field while avoiding substantial contamination from broader but less directly related post-growth

debates.

Across these robustness exercises, the main conclusions remain unchanged: the intellectual foundations, conceptual evolution, and emerging research fronts identified in the analysis are not driven by a particular threshold, counting convention, or corpus definition, but reflect persistent structures in the underlying literature.

4 Discussion

The three science-mapping analyses reveal that the evolution of degrowth has been characterised less by the replacement of one research agenda with another than by a gradual reorganisation of the field's intellectual centre. The defining shift is that the literature has moved from establishing the conceptual case for degrowth toward empirically assessing the environmental limits of continued economic growth and the conditions under which post-growth transitions might be realised. Around this shift, degrowth has developed many of the characteristics of a mature research field: a stable knowledge base, a differentiated conceptual vocabulary, and an increasingly specialised research frontier.

On the first question of the knowledge base, the co-citation results point less to novelty than to consolidation. Degrowth rests on a small, stable canon, and the existence of such a shared foundation is itself a hallmark of disciplinary maturity. A concentrated citation core alone, however, is not sufficient evidence of intellectual coherence. Rapidly expanding fields often develop stable citation patterns through preferential-attachment dynamics, even when the underlying scholarship remains conceptually fragmented. What distinguishes the degrowth literature is not simply the existence of a citation core, but its internal organization. Rather than a loose collection of highly cited works, the core forms a coherent intellectual structure whose consolidation extends beyond the three-cluster partition. The clusters are not separate schools but parts of one connected core: works in different clusters are still heavily co-cited with one another rather than forming self-contained groups (Appendix A.1). The fact that these traditions continue to cite one another extensively suggests that degrowth has developed a shared intellectual space rather than a set of isolated schools.

The applied post-growth and green-growth-critique cluster now forms the canon's densest cluster (its references share the highest mean internal association strength) and, once co-citation is recomputed period by period, its most rapidly expanding one: its share of the knowledge base rises from a marginal presence before 2011 to 43% by 2021–2025, overtaking the foundational core, whose share more than halves over the same span (Appendix A.2). This expansion indicates a reconfiguration of the field's intellectual structure: the centre of gravity has gradually shifted from primarily articulating degrowth as an alternative vision toward empirically examining whether technological, institutional, and policy-based strategies can deliver sufficient and sustained reductions in environmental pressures, while intensifying the debate over the feasibility of green growth (Hickel and Kallis, 2020; Haberl et al., 2020). The foundational core of degrowth and ecological economics, as well as the political-ecology and post-development strand, remain important intellectual anchors, but the field's recent momentum has shifted toward this empirical front.

On the second question, the conceptual space, the co-word maps show the field's vocabulary being transformed, not merely growing. The most revealing result is the starting point: before 2000 the term “degrowth” was not used as an economic concept within the indexed literature but primarily as a bio-

Table 3: The degrowth field across three lenses: foundations, vocabulary, and frontier

Thematic stream	Knowledge base (co-citation)	Conceptual space, 2021–2025 (co-word)	Research front, 2021–2025 (coupling)
Ecological-economic foundations & biophysical limits	Cluster 1: foundational theory and ecological economics	Blue, orange: sustainability, resource governance, steady-state thinking, and planetary boundaries	Red: post-growth macroeconomic and climate-feasibility modelling
Political ecology, post-development, & decolonial perspectives	Cluster 2: political ecology, post-development, and degrowth vocabulary	Green: political ecology, global justice, and environmental/climate justice	Orange: decolonial degrowth, political ecology, and global justice
Applied post-growth research & the critique of green growth	Cluster 3: applied post-growth research and the critique of green growth	Red, purple, pink: energy–climate transitions, socio-ecological transformation, and alternative economic systems	Green: circular-economy and green-growth research against a critical, degrowth-informed alternative
Sectoral & applied transitions	Limited presence as a distinct foundational stream	Brown, grey: tourism, urban planning, and eco-social transitions	Purple, brown, blue, light-blue, teal, grey: urban transformation, tourism, business innovation, and public attitudes toward growth
Justice, well-being, & provisioning	Embedded within Clusters 2–3	Yellow, teal: social justice, well-being, post-development, and economic systems	Pink, yellow: eco-social policy, welfare state, needs, and sufficiency

Notes: Each row traces one thematic stream across the three science-mapping lenses: its anchoring works in the co-citation knowledge base, its vocabulary in the 2021–2025 co-word map, and its position on the 2021–2025 coupling research front. Cell entries name the relevant clusters from Figures 4, 8, and 10.

logical term. The socio-economic field did not capture the word so much as repurpose it, emerging in the mid-2000s alongside a physiological usage that persists in regeneration biology to this day; what changed is which meaning came to dominate the literature. Once repurposed, the vocabulary changed in kind, not merely in size. The field did not simply accumulate more publications around the same critique. It developed an increasingly differentiated conceptual architecture linking ecological limits, justice, governance, provisioning, urban transformation, and post-development. This change in the kind of vocabulary, more than the rise in output, is the clearest sign that degrowth has become a field of ideas in its own right.

On the third question, the research front, bibliographic coupling shows the field undergoing structured diversification. Whereas the early-2010s research front remained centred around a relatively small number of foundational debates, the recent frontier has expanded into empirical and sector-specific niches: it runs at once toward operationalisation (modelling, provisioning, post-growth business, urban policy) and toward a globally and decolonially situated discourse. This asymmetry is visible directly in Table 3: the sectoral and applied strand, barely present as a distinct stream in the knowledge base, fragments into six separate clusters at the research front, the clearest sign that the field has become increasingly differentiated. Viewed together, Table 3 shows that the three science-mapping techniques recover the same broad thematic streams at different analytical levels. The intellectual foundations identified by co-citation reappear in the conceptual vocabulary and again at the contemporary research frontier, providing a coherent account of the field’s evolution despite each method relying on a different relational structure.global.

These convergent movements carry three implications. Theoretically, degrowth now has the stable canon and shared vocabulary of a maturing field, even as its centre of gravity moves from foundational debate toward applied, sector-specific research. Methodologically, the breadth of the corpus shows that degrowth is no longer the preserve of ecological economics and political ecology but a genuinely cross-disciplinary domain spanning business, energy, urban studies, and tourism. For policy, the turn toward modelling, governance, and provisioning marks a field increasingly able to inform concrete post-growth strategies. Yet the same evidence also identifies the field’s unresolved challenge. While degrowth scholarship has become increasingly sophisticated in diagnosing ecological limits and developing alternative economic imaginaries, the institutional mechanisms through which such transitions could be politically achieved remain comparatively underdeveloped. Questions of governance, coalition-building, and distributional conflict therefore remain important unresolved dimensions of the degrowth research agenda, echoing earlier calls to place the politics of transition at the centre of degrowth scholarship ([Hanaček et al., 2020](#)).

At the same time, this interpretation requires caution. The narrative arc from a young and scattered literature to a consolidated, applied, and policy-relevant field is also the type of trajectory that bibliometric self-histories of research domains tend to produce, since expanding literatures mechanically accumulate more clusters, citations, and thematic breadth over time. What supports interpreting the pattern identified here as more than a generic growth trajectory is the convergence of independently computed structures: co-citation (shared references), co-word analysis (shared vocabulary), and bibliographic coupling (shared reference lists) rely on different relational signals, yet they point to the same underlying transformation. This convergence is not mechanically implied by publication growth alone.

5 Conclusion

This paper has mapped the intellectual landscape of degrowth through a bibliometric analysis of 2,927 peer-reviewed publications (1968–2025), combining co-citation, co-word, and bibliographic-coupling analyses. Its central finding is a structural one: the field’s intellectual centre of gravity has shifted from primarily articulating degrowth as an alternative to growth toward increasing empirical engagement with the environmental feasibility of green growth and the institutional conditions for post-growth transitions. Around that shift, degrowth emerges as a field that has matured along three axes: from conceptual consolidation to thematic diversification, from critique to operationalisation, and from a predominantly European intellectual tradition toward a more globally attentive and increasingly decolonial research agenda. By spanning the full disciplinary range of the field, the study offers the most extensive quantitative mapping of degrowth scholarship to date and complements its many discipline-bound reviews with a macro-level, relational perspective on its past, present, and emerging directions.

The analysis is subject to several limitations. Its corpus is defined by a title- and abstract-level query for the term “degrowth,” which may omit cognate work framed as post-growth, steady-state, or *décroissance* without using the keyword. Abstract coverage is incomplete for older and non-English records, which constrains the co-word analysis for the earliest period. Although the clustering procedures are transparent and robust to alternative specifications, they remain dependent on choices regarding thresholds, network construction, and bibliographic disambiguation.

These limitations point toward future work. Subsequent studies could broaden the query to the wider

post-growth lexicon and complement the present macro-level map with focused, qualitative readings of the emerging niches identified here, particularly post-growth macroeconomic modelling, decolonial degrowth, urban transformation, and post-growth business. Whether degrowth ultimately becomes an enduring scientific paradigm or remains one perspective within sustainability science will depend less on further conceptual debate than on its ability to generate empirically grounded knowledge about the governance of socio-ecological transitions.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI's ChatGPT and Anthropic's Claude to assist with language editing, improving clarity and readability, and refining the presentation of the manuscript, as well as with writing and debugging R code used to implement the bibliometric analysis. These AI tools were not used to determine the study's research questions, methodological design, or the interpretation of results and conclusions, which remain the authors' own. After using these tools, the authors carefully reviewed and edited all content and code and take full responsibility for the accuracy, integrity, and final content of this article.

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A Robustness Appendix

This appendix reports the full robustness battery summarised in Section 3.4. Each check re-estimates the relevant network or ranking from the corpus used in the main text, perturbing one analytical choice at a time and comparing the result with the reported solution; a final check (Appendix A.6) instead assesses the language coverage of the keyword-defined corpus. Throughout, the adjusted Rand index (ARI) measures the agreement between a perturbed partition and the partition reported in the paper, with 1.00 denoting an identical clustering; the cluster count records communities with at least three nodes (co-citation and coupling) or two nodes (co-word).

A.1 Co-citation knowledge base

We perturb the co-citation map along six dimensions. Table A.1 reports its sensitivity to the number of references retained as nodes: the three-cluster solution holds across the entire range tested, from seventy to one hundred and twenty most-cited references. Table A.2 reports the sensitivity of the partition itself, holding the ninety-node network fixed. The reported resolution coincides with the modularity maximum ($Q = 0.113$ at $\gamma = 1.0$);¹¹ the Louvain algorithm (Blondel et al., 2008) recovers the identical partition, whereas the agglomerative Walktrap (Pons and Latapy, 2006) and fast-greedy (Clauset et al., 2004) methods return a coarser two-way split that merges two of the three clusters without cutting across them. The solution survives fifty random seeds and does not depend on the manual de-duplication of cited-reference records.

Table A.1: Co-citation map: sensitivity to the node cutoff

Cutoff (refs)	Nodes	Clusters	Cluster sizes	ARI vs. reported
70	70	3	30, 27, 13	0.90
80	80	3	34, 32, 14	0.88
85	85	3	34, 34, 17	0.88
90 (reported)	90	3	36, 31, 23	–
100	100	3	39, 37, 24	1.00
110	110	3	43, 40, 27	0.97
120	120	3	46, 44, 30	0.75

Notes: The co-citation network is re-estimated keeping the most-cited references down to each cutoff. Cluster sizes are node counts, and the ARI is computed against the reported ninety-node solution. The three-cluster structure holds across the full range tested, from seventy to one hundred and twenty references.

¹¹The modularity Q is a standard measure of how cleanly a network separates into clusters: it compares the share of link weight falling *within* clusters to what a random wiring of the same nodes would produce, so higher values mean more clearly separated clusters. The resolution γ is the parameter of the Leiden algorithm that sets how finely the network is divided, with larger values yielding more and smaller clusters (both are defined in Section 2).

Table A.2: Co-citation map: sensitivity of the partition (ninety nodes)

Perturbation	Clusters	ARI vs. reported
Resolution at the modularity maximum ($\gamma = 1.0$, $Q = 0.113$)	3	1.00
Louvain algorithm	3	1.00
Walktrap / fast-greedy	2	0.55
Association-strength edge weighting	3	0.68
Random seed (modal over 50 seeds)	3 (42/50)	0.91 (mean)
Raw references (manual de-duplication off)	3	0.85

Notes: Q is the modularity (higher means more clearly separated clusters) and γ the Leiden resolution (higher yields more, smaller clusters). Walktrap and fast-greedy merge two of the three clusters into one without splitting any reported cluster across communities; the seed and de-duplication rows report, respectively, the mean pairwise ARI across fifty seeds and the ARI on shared nodes between the canonical and raw-reference networks.

These checks establish that cluster membership and the number of clusters are stable. The thematic labels assigned to each cluster (e.g., “applied post-growth research and the critique of green growth” and “foundational theory”) remain qualitative interpretations of the clusters’ most prominent terms and references rather than outputs directly produced by the algorithm.

A.2 Temporal co-citation: migration of the knowledge base

The co-citation map in the main text aggregates the whole period (1968–2025) and so cannot, on its own, show when a cluster rose to prominence. To make the migration explicit, we hold the ninety nodes and their three-cluster assignment fixed from the full-corpus map and recompute co-citation using only the documents published in each period. For period t we restrict to corpus documents with publication year in t , count co-citations among the fixed nodes, and give each cluster its share of the period’s total node strength,

$$\text{share}_c(t) = \sum_{i \in c} s_i(t) / \sum_i s_i(t),$$

where $s_i(t)$ is reference i ’s co-citation strength in period t . Because the cluster identities are fixed, the shares are directly comparable across periods without any cross-period matching. Table 2 reports the result. The foundational cluster supplies almost the entire knowledge base in 2001–2010 (98.2%) and then declines steadily (63.1%, then 31.2%), while the applied post-growth and green-growth-critique cluster rises from absence (0.0%) through 12.0% to 43.1%, overtaking the foundational core to become the largest share by 2021–2025. The migration is therefore a property of the data, not of the static layout: the green-growth share grows by 12.0 percentage points across the 2010s and by a further 31.1 across the 2020s. The earliest period (1968–2000) contributes no co-citations among the retained nodes, consistent with the field’s knowledge base not yet existing.

A.3 Conceptual structure (co-word)

For each of the four periods, Table A.3 varies the term-frequency threshold around its reported value and reports the resulting cluster count, together with the seed stability at the reported threshold. The number of thematic clusters is preserved in every period as the threshold moves. Seed stability is highest in the

two sparse early periods, where the partition is identical across all thirty seeds, and remains modally stable in the denser recent periods; exact term placement is more sensitive there, consistent with reading these maps at the level of broad thematic formations.

Table A.3: Co-word maps: sensitivity to the frequency threshold and the random seed

Period	Threshold	Terms	Clusters	ARI / seed stability
1968–2000	2	312	3	ARI 0.49
	3 (reported)	117	3	modal 3 (30/30), mean ARI 0.89
	4	56	3	ARI 0.60
2001–2010	3	205	4	ARI 0.48
	4 (reported)	94	4	modal 4 (30/30), mean ARI 0.79
	5	52	4	ARI 0.61
2011–2020	9	98	8	ARI 0.52
	12 (reported)	53	7	modal 8 (20/30), mean ARI 0.65
	15	39	8	ARI 0.44
2021–2025	14	128	9	ARI 0.57
	18 (reported)	85	10	modal 10 (17/30), mean ARI 0.66
	22	61	10	ARI 0.76

Notes: Clusters count communities with at least two nodes. For the reported threshold the cell gives the modal cluster count across thirty seeds (with the number of seeds attaining it) and the mean pairwise ARI against the first seed; for the alternative thresholds it gives the ARI on the terms shared with the reported network.

A.4 Research front (bibliographic coupling)

Table A.4 varies the citation-count entry threshold around its reported value for each coupling window and reports the seed stability at the reported threshold. The cluster count moves by at most one or two documents-worth of structure under either perturbation, and the modal partition is stable across seeds, supporting a thematic rather than node-by-node reading of the research front.

Table A.4: Coupling maps: sensitivity to the citation threshold and the random seed

Window	Citation threshold	Documents	Clusters	ARI / seed stability
2011–2020	≥ 20	300	8	ARI 0.54
	≥ 30 (reported)	241	8	modal 8 (22/30), mean ARI 0.78
	≥ 40	198	9	ARI 0.60
2021–2025	≥ 8	322	11	ARI 0.68
	≥ 12 (reported)	243	11	modal 11 (16/30), mean ARI 0.84
	≥ 16	197	12	ARI 0.81

Notes: Clusters count communities with at least three nodes. For the reported threshold the cell gives the modal cluster count across thirty seeds (with the number of seeds attaining it) and the mean pairwise ARI against the first seed; for the alternative thresholds it gives the ARI on the documents shared with the reported network.

A.5 Counting conventions

Table A.5 re-estimates the two descriptive rankings most exposed to the choice of counting convention. The geographic distribution is almost unchanged whether countries are credited by corresponding author,

as reported in the paper, or by whole counting across all co-author affiliations. Author productivity retains the same prolific core under full and fractional counting; fractional counting, which discounts authorship on heavily co-authored papers, reorders the mid-list but leaves the leading authors in place.

Table A.5: Descriptive rankings: sensitivity to the counting convention

Ranking	Baseline vs. alternative	Spearman ρ	Top overlap
Country output	Corresponding author vs. all-author whole counting	0.95	12/12
Author productivity	Full vs. fractional ($1/n$) counting	0.52	10/15

Notes: Spearman ρ is the rank correlation between the baseline and alternative counts across all shared units (eighty countries; 3,727 authors). Top overlap is the number of shared entries among the top twelve countries and the top fifteen authors, respectively.

A.6 Language coverage

The corpus is built from a title-and-abstract search for the English term *degrowth* (Section 2); this is a search term, not an author-supplied keyword, and it excludes degrowth scholarship published under cognate terms in other languages. We gauge the size of that gap with OpenAlex. Searching each cognate term directly is misleading, because words such as *décroissance*, *decrecimiento*, and *decrescita* are everyday words for “decrease” and return large volumes of unrelated work (radioactive decay, eigenvalue decrease, demographic decline). We therefore restrict each term to OpenAlex’s language-independent “Degrowth” subject concept, which keeps only works it classifies as concerned with degrowth and discards the everyday decrease/decay uses.¹²

Table A.6 reports the result: the genuinely degrowth-related literature carrying a non-English term is small, under twenty documents per language and roughly sixty in total, a few percent of the corpus. Excluding it therefore does not materially affect the field’s aggregate structure. It does, however, bear on the field’s early intellectual history, since the foundational *décroissance* works, beginning with [Grinevald and Rens \(1979\)](#), predate and fall outside the keyword corpus; we therefore cite them directly in the narrative rather than recover them from the data.

Table A.6: Language coverage: degrowth literature by search term and concept

Term (language)	Raw hits	Degrowth-concept
<i>degrowth</i> (English)	2,974	1,758
<i>décroissance</i> (French)	2,575	18
<i>decrecimiento</i> (Spanish)	1,580	16
<i>Postwachstum</i> (German)	112	13
<i>decrecemento</i> (Portuguese)	386	10
<i>decrescita</i> (Italian)	142	4
<i>decreixement</i> (Catalan)	52	1

Notes: “Raw hits” is an OpenAlex title-and-abstract search for the term; “Degrowth-concept” restricts it to works also carrying OpenAlex’s language-independent “Degrowth” concept.

¹²OpenAlex tags each work with subject concepts; restricting the search to its “Degrowth” concept keeps only works classified as about degrowth and discards the everyday “decrease/decay” and biological-shrinkage uses.

A.7 Keyword-boundary sensitivity

A second concern about the corpus boundary is that the English keyword *degrowth* may miss adjacent post-growth scholarship framed in other words, such as post-growth, steady-state economics, a-growth, or doughnut economics. We assess this with OpenAlex. Searching the cognate terms directly is uninformative, because the searches are dominated by false positives unrelated to the field: a title/abstract search for “post-growth” returns more than 250,000 records, most of them developmental-biology uses of the phrase, and “a-growth” matches millions of documents containing the words “a growth”. We therefore again use OpenAlex’s language-independent “Degrowth” concept to isolate the genuinely degrowth-related works carrying each cognate term. Table A.7 reports the result: once disambiguated, each cognate framing is small (a few dozen to roughly two hundred degrowth-concept works). More directly, of the 2,183 works OpenAlex assigns to the Degrowth concept, 1,980 (91%) contain the word *degrowth* in their title or abstract, so only about 200 degrowth-related works (9% of the concept) would be added by broadening the query, too few to alter the field’s aggregate structure. A further reason the boundary matters little is structural: the co-citation knowledge base is built from the references *cited by* the corpus, so post-growth work that builds on the same foundational texts is already reflected in the co-citation structure even when it falls outside the keyword.

Table A.7: Keyword-boundary sensitivity: adjacent post-growth framings

Term	Raw hits	Degrowth-concept
<i>degrowth</i> (reference)	3,242	1,980
post-growth	255,838	194
steady-state economy	9,369	47
prosperity without growth	1,869	21
wellbeing economy	6,125	36
doughnut economics	240	12

Notes: OpenAlex title-and-abstract searches, restricted to articles, reviews, books, and book chapters published up to 2025. “Raw hits” counts every record matching the term and is dominated by unrelated uses (for example, “post-growth” in developmental biology); “Degrowth-concept” restricts the same search to works carrying OpenAlex’s language-independent “Degrowth” concept. “a-growth” is omitted because the search matches the ordinary phrase “a growth” in millions of records and cannot be disambiguated by counts alone.